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1. System Bus Specifications

The system bus must support:

- **Address lines**: 16 (since all microprocessors use a 16-bit address bus).
- **Data lines**:
- The **16-bit processor** will use a full 16-bit data bus.
- Each **8-bit processor** will use only 8 out of the 16 data lines.
- **Control lines**:
- **Interrupt lines**: 4 (since the system supports a maximum of 4 interrupts).
- **DMA lines**: 1 (DMAREQ, DMAACK).
- **Bus arbitration signals**: HOLD, HLDA (to manage multiple processors on the bus).

2. Bus Communication Protocol

The communication between devices follows a structured protocol:

Read/Write Cycle:

- Processor places the address on the address bus.
- Processor sets Read/Write control lines (R/W = 1 for read, R/W = 0 for write).
- Memory or I/O responds by placing data on the data bus (for read) or storing incoming data (for write).
- Acknowledgment signal is sent when data transfer is complete.

Interrupt Handling:

- Each processor can request an interrupt via one of the 4 available interrupt lines (INTR0–INTR3).

- The main processor (16-bit) will handle priority and service the request using the INTRACK signal.
- The processor executes the interrupt service routine (ISR) before resuming normal execution.

DMA (Direct Memory Access) Cycle:

- A device requests the bus using the DMAREQ signal.
- The processor acknowledges using the DMAACK signal and temporarily gives control of the bus.
- The DMA controller transfers data between memory and the device without processor intervention.
- Once the transfer is complete, control returns to the processor.

3. Device Interfacing with the System Bus

- The **Bus Arbitration Logic** ensures that only one processor accesses the bus at a time.
- **Memory (64 KB x 2 = 128 KB)** is shared using the full 16-bit address space.
- **8-bit processors** only use the lower half (8 bits) of the data bus, so additional logic is needed to enable their access.
- **I/O controllers** communicate through the bus and can use interrupt-driven or programmed I/O.
- **DMA controller** bypasses the processor for high-speed data transfer.

Final Thoughts

Your system needs proper bus arbitration to avoid conflicts between processors, memory, and DMA. You might also consider a priority mechanism for interrupt handling.